Introduction to Software Engineering

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Software Requirements Specification

UniLib — University Library Management System

Team T05 · 2 girls 2 cúp | Baseline v1.0.0-frozen | November 2026

Objectives

This document focus on the following topics:

- Completing the Software Requirements Specification (SRS) document with the following sections:
- Problem Statement
- Stakeholders
- Functional, non-functional and domain requirements
- Use case model and use case specifications
- Prototype / mockups
- Traceability and requirements review
- Including an AI usage log for this phase
- Understanding the Software Requirements Specification (SRS) document.

Member Contribution Assessment

Member contribution is distributed equally at 25.0% per member (100.0% total) in accordance with Team Contract §3.1 and §3.5. Evidence is linked to repository tasks, issues, and merged pull requests:

ID	Name	Role(s)	Contribution (%)	Evidence
23127520	Nguyễn Phan Thế Vinh	Team lead / PM, Tester	25%	Tasks PA2-V1–V6: Chaired PA2 kickoff meeting, authored kickoff minutes (2026-10-23-kickoff.md); established and froze Master ID Registry (00-id-registry.md); drafted Acceptance Criteria (AC-F01-01 to AC-F07-02); authored Requirements Traceability Matrix (RTM) baseline; coordinated formal requirements inspection walkthrough; compiled and finalized PA2 docx/pdf. GitHub Issues: #31, #32; Merged PRs: #82, #83, #84, #86.
24127469	Nguyễn Thanh Thảo Nguyên	Business analyst	25%	Tasks PA2-N1–N6: Formulated preliminary BA inquiries and domain glossary (02-glossary.md); authored Purpose, Scope, Problem Statement, and Stakeholder analysis; detailed all 32 Functional Requirements (F01–F07) and 10 Business Rules (BR-01 to BR-10); constructed Use Case Model (diagram, actors, UC-01 to UC-12 list) and authored full Use Case specifications. GitHub Issues: #20, #21, #22, #23, #24, #25, #26, #27; Merged PRs: #82, #83, #84.
24127324	Vũ Thị Ánh	Designer / Architect	25%	Tasks PA2-A1–A6: Proposed screen inventory (SCR-01 to SCR-12) and UI layout hierarchy; authored Non-Functional Requirements (NFR-01 to NFR-08); designed low-fidelity wireframes for Reader and Librarian/Admin circulation desks; built interactive UI prototype (Figma); performed UI/UX accessibility review. GitHub Issues: #28, #29, #30; Merged PRs: #77, #79, #86.
24127228	Trần Hà Anh Quân	Developer	25%	Tasks PA2-Q1–Q6: Validated repository skeleton and technical stack feasibility; analyzed PostgreSQL concurrency, ACID transaction requirements, and row-level locking for circulation states (pa2-quan-technical-review.md); reviewed NFR metrics (load, latency, security) and technical deliverability for PA3/PA5. GitHub Issues: #28, #31; Merged PRs: #2, #75.

Introduction

Purpose and Scope

This Software Requirements Specification (SRS) document details the complete functional, non-functional, and domain requirements for the UniLib (University Library Management System) web application. The primary purpose is to establish an unambiguous contractual baseline between stakeholders and the development team (T05) for subsequent architecture design (PA3), test planning (PA4), and full implementation (PA5).

The product scope encompasses seven core features: six Must-have features committed for full delivery in PA5—F01 Authentication & Role-Based Access Control, F02 Book Catalog Management, F03 Book Copy Inventory Management, F04 Book Search & Browse, F05 Borrow & Return Management (Circulation Desk), and F06 Book Reservation—along with one prioritized Should-have feature: F07 Reader Overdue & Loan Monitoring Dashboard.

Strictly out of scope for UniLib are: e-book reading and digital rights management (DRM), real online payment gateways or fine transactions, physical RFID or hardware barcode scanners (copy barcodes are input via UI/virtual entry), inter-library loans, AI recommendation engines, production SMS/email gateway integrations (all status alerts are provided in-app), and native mobile applications.

The intended readers of this document are course lecturers, teaching assistants (TAs), project evaluators, and all team members responsible for software design, implementation, and quality assurance.

Glossary

Term	Definition
Book Title	A bibliographic entity representing a published literary or academic work, uniquely identified by ISBN, title, author(s), category, publisher, and publication year.
Book Copy	A specific physical inventory unit of a Book Title possessed by the library, uniquely tracked by Barcode / Copy ID, shelf location, and lifecycle status (Available, Borrowed, Reserved, Damaged/Lost).
Loan	A borrowing transaction linking an active Reader, a physical Book Copy, a checkout timestamp, a calculated due date (14 calendar days), and a loan status (Active, Returned, Overdue).
Reservation (Hold)	A title-level request placed by an eligible Reader when zero copies are Available, queued in strict FIFO order and held for a 72-hour pickup window upon return of a copy.

Overdue Loan	A loan record whose borrowed copy has not been returned or renewed on or before the designated due date, triggering automated borrowing suspension.
ID Registry	The master identification baseline (docs/pa2/00-id-registry.md) freezing all identifiers (F01–F07, FR-F0x-nn, NFR-nn, BR-nn, UC-01–12, SCR-01–12, AC-F0x-nn) for parallel development.

Problem Statement

University and departmental libraries frequently manage collections of thousands of academic volumes, journals, and reference materials. In many institutions, library operations still rely on fragmented spreadsheets, paper logbooks, or disconnected desktop software. This operational fragmentation creates critical bottlenecks: patrons cannot easily discover which titles are available on shelf without physically visiting the circulation desk; librarians struggle to reconcile physical copies with circulation records; and overdue loans are frequently overlooked, leading to unreturned books and resource scarcity.

UniLib addresses these challenges by delivering a centralized, responsive web-based library management platform. The system automates the core circulation lifecycle—cataloging, copy inventory control, patron checkout and check-in, due date calculations, overdue tracking, and a fair First-In, First-Out (FIFO) reservation queue for high-demand titles.

Operating Environment:
• Client: Modern evergreen web browsers (Google Chrome 120+, Mozilla Firefox 120+, Microsoft Edge 120+, Safari 17+) supporting HTML5, CSS3, and JavaScript ES2022 on desktop and tablet viewport widths (1024px to 1920px).
• Server & Database: Node.js 22 LTS runtime with Express REST API, deployed on Render; PostgreSQL 16 relational database with ACID transactions and row-level locking for concurrency control; React frontend hosted on Vercel.

Design and Implementation Constraints:
• Standardized technology stack: React + Vite on frontend; Express + Node.js on backend; PostgreSQL relational database.
• Scope boundaries strictly capped at 6 Must-have features (F01–F06) and 1 Should-have feature (F07).
• Physical barcode scanners and external SMS/email infrastructure are omitted; all barcode operations utilize virtual UI inputs and all notifications are delivered via responsive in-app status dashboards.

Stakeholders

The following stakeholder groups are identified for the UniLib system, detailing their primary operational roles and core concerns:

#	Stakeholder	Description	Interests / concerns
1	Project Sponsor & Evaluator	Course Instructor and Teaching Assistants (FIT - HCMUS)	Verification of software engineering rigor, adherence to Waterfall process gates, strict traceability from requirements to code/tests, and academic integrity.
2	Reader / Student Patron	University students, researchers, and faculty members using library resources	Rapid keyword search, real-time copy availability visibility, fair FIFO book reservations, clear personal loan dashboard, and timely overdue notices.
3	Librarian / Circulation Staff	Library operations personnel managing collection and circulation desk	Fast and accurate book checkout/check-in, automated due date and overdue calculations, physical copy inventory tracking, and reservation hold processing.
4	System Administrator	IT administrators managing library system configuration and user accounts	Role-based access control (RBAC), user account lifecycle management, robust audit trails, data integrity, and high system availability.

Requirements

Functional Requirements

All 32 Functional Requirements are categorized under features F01–F07 and prioritized using MoSCoW, strictly following the frozen Master ID Registry (pa/PA2/00-id-registry.md):

ID	Requirement	Feature	Priority	Source
FR-F01-01	The system shall authenticate users using username/email and password credentials.	F01	Must	PA0 §2.4, PA1 Baseline
FR-F01-02	The system shall generate a secure JWT upon authentication and enforce role-based access control (Reader, Librarian, Administrator).	F01	Must	PA0 §2.4, PA1 Baseline
FR-F01-03	The system shall allow authenticated users to log out and invalidate the client-side session token.	F01	Must	PA1 Baseline
FR-F01-04	The system shall provide Administrator capabilities to create user accounts, assign roles, and toggle account active status.	F01	Must	PA0 §2.3, PA1 Baseline
FR-F02-01	The system shall allow Librarians to create new book title records with ISBN, Title, Author(s), Category, Publisher, and Publish Year.	F02	Must	PA0 §2.4, PA1 Baseline
FR-F02-02	The system shall allow Librarians to edit metadata of existing book titles.	F02	Must	PA0 §2.4, PA1 Baseline
FR-F02-03	The system shall allow Librarians to delete/deactivate a book title only when no physical copies or active loans/holds are linked to it.	F02	Must	PA1 Baseline, BR-09
FR-F02-04	The system shall display aggregated copy statistics (Total, Available, Borrowed, Reserved) on the book title detail view.	F02	Must	PA1 Baseline
FR-F03-01	The system shall allow Librarians to register physical copies with unique Barcode/Copy ID and Shelf Location.	F03	Must	PA0 §2.4, PA1 Baseline
FR-F03-02	The system shall allow Librarians to update physical copy metadata including shelf location and physical condition notes.	F03	Must	PA1 Baseline
FR-F03-03	The system shall track and transition copy lifecycle states: Available, Borrowed, Reserved, and Damaged/Lost.	F03	Must	PA0 §2.4, PA1 Baseline
FR-F03-04	The system shall allow Librarians to decommission a physical copy only when current status is Available or Damaged/Lost.	F03	Must	PA1 Baseline, BR-09
FR-F04-01	The system shall allow users to execute keyword searches across Title, Author, Category, and ISBN.	F04	Must	PA0 §2.4, PA1 Baseline
FR-F04-02	The system shall allow users to filter search results by Category, Availability Status, and sort by Title or Publish Year.	F04	Must	PA1 Baseline

FR-F04-03	The system shall display real-time physical copy availability counts and shelf locations on search results and book detail views.	F04	Must	PA0 §2.4, PA1 Baseline
FR-F04-04	The system shall support server-side paginated browsing of the catalog with a default page size of 10 items.	F04	Must	PA1 Baseline
FR-F05-01	The system shall allow Librarians to issue a book loan (checkout) by selecting an active Reader ID and an Available Copy ID.	F05	Must	PA0 §2.4, PA1 Baseline
FR-F05-02	The system shall validate borrowing eligibility before checkout: Reader active status, < 3 active loans (BR-01), and 0 overdue loans (BR-04).	F05	Must	Kickoff Decision D-PA2-01
FR-F05-03	The system shall compute the loan due date exactly 14 calendar days from checkout date (BR-02) and update copy status to Borrowed.	F05	Must	Kickoff Decision D-PA2-01
FR-F05-04	The system shall allow Librarians to process book returns (check-in), recording return timestamp and updating loan status to Returned.	F05	Must	PA0 §2.4, PA1 Baseline
FR-F05-05	Upon return processing, if pending reservations exist, the system shall transition the copy to Reserved for the top FIFO reader; else to Available.	F05	Must	PA0 §2.4, BR-08
FR-F05-06	The system shall allow loan renewal by 14 days once if the loan is not overdue and no reservations are pending for the title (BR-03).	F05	Must	Kickoff Decision D-PA2-01
FR-F05-07	The system shall automatically detect and flag overdue loans when current date exceeds the calculated due date.	F05	Must	PA0 §2.4, PA1 Baseline
FR-F06-01	The system shall allow an eligible Reader to reserve a book title when zero copies are in Available status (BR-05).	F06	Must	PA0 §2.4, PA1 Baseline
FR-F06-02	The system shall enforce reservation limits: <= 2 active reservations per reader (BR-06) and no active loan for the same title.	F06	Must	PA1 Baseline, BR-06
FR-F06-03	The system shall maintain an automated First-In, First-Out (FIFO) queue for reservations per book title (BR-07).	F06	Must	PA0 §2.4, BR-07
FR-F06-04	When a copy transitions to Reserved, the system shall hold the copy for a 72-hour pickup window (BR-08).	F06	Must	Kickoff Decision D-PA2-02
FR-F06-05	The system shall allow Readers to cancel reservations, and auto-expire uncollected holds after 72 hours, advancing the FIFO queue.	F06	Must	Kickoff Decision D-PA2-02
FR-F06-06	The system shall fulfill a reservation when the holding reader checks out the held copy at the circulation desk.	F06	Must	PA0 §2.4, PA1 Baseline
FR-F07-01	The system shall provide Readers a personal dashboard displaying active loans, due dates, renewal actions, and active reservations.	F07	Should	PA1 Scope §2.4
FR-F07-02	The system shall display prominent overdue warnings and borrowing restriction alerts on the Reader dashboard.	F07	Should	PA1 Scope §2.4
FR-F07-03	The system shall provide Librarians an overdue monitoring console listing all overdue loans, patron contact details, and days overdue.	F07	Should	PA1 Scope §2.4

Non-Functional Requirements

The following measurable Non-Functional Requirements define performance benchmarks, security standards, usability criteria, and system quality attributes:

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	Under 100 concurrent simulated users, catalog search queries shall return response payloads in <= 1.0 s at P95 across 10,000 book records.	Automated Load Test (k6 / Artillery)
NFR-02	Performance	Initial application load time shall achieve Largest Contentful Paint (LCP) <= 2.5 s at P90 on standard broadband/4G profiles.	Lighthouse Performance Audit
NFR-03	Security	User passwords shall be hashed using bcrypt with salt rounds >= 10; plain-text credentials shall never be logged or transmitted in response payloads.	Code Review & Unit Test Inspection
NFR-04	Security	All protected REST endpoints shall validate JWT claims; unauthenticated requests return HTTP 401; unauthorized roles return HTTP 403.	Automated API Integration Tests (Jest + Supertest)
NFR-05	Data Integrity	Circulation state changes (checkout, return, hold allocation) shall execute within atomic DB transactions with row-level locks, guaranteeing 0 race conditions.	Concurrency Integration Test Suite
NFR-06	Usability	The user interface shall achieve Google Lighthouse Accessibility score >= 85, color contrast ratio >= 4.5:1, and support keyboard navigation.	Lighthouse Accessibility Audit & Manual Walkthrough
NFR-07	Compatibility	The web application shall render correctly on modern evergreen browsers (Chrome 120+, Firefox 120+, Edge 120+, Safari 17+) on viewports 1024px to 1920px.	Cross-browser Verification & Viewport Emulation
NFR-08	Reliability	The system shall display user-friendly error banners upon network/API failures, and support database schema restoration via seed scripts in < 5 minutes.	Fault Injection Test & Recovery Run

Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	Maximum Active Loan Limit: A Reader is strictly limited to a maximum of 3 concurrently active (unreturned) book loans across the entire library system.	Library Circulation Policy §2.1 (Kickoff 23/10)
BR-02	Standard Loan Period: The default borrowing duration is 14 calendar days from checkout timestamp. Due date is calculated automatically at checkout.	Academic Calendar & Loan Rules §3.2 (Kickoff 23/10)
BR-03	Loan Renewal Policy: A loan may be renewed exactly once for an additional 14 calendar days, provided the loan is not overdue and has 0 title reservations.	Circulation Renewal Policy §3.4

BR-04	Overdue Borrowing Suspension: Any Reader possessing at least one overdue loan is immediately blocked from new checkouts and new reservations until overdue items are returned.	Circulation Sanction Rule §4.1
BR-05	Reservation Eligibility: A Reader may place a reservation for a book title if and only if zero physical copies are currently Available, and reader has no active loan of the title.	Reservation Policy §5.1
BR-06	Maximum Active Reservations: A Reader may hold a maximum of 2 active reservations concurrently across the system.	Reservation Policy §5.2
BR-07	Reservation Priority Order: Book reservations are queued and processed in strict First-In, First-Out (FIFO) chronological sequence based on reservation timestamp.	Reservation Queue Protocol §5.3
BR-08	Reservation Hold Pickup Window: When a returned copy is routed to fulfill a reservation, it is held with status Reserved for exactly 3 calendar days (72 hours) before auto-expiry.	Hold Shelf Policy §5.4 (Kickoff 23/10)
BR-09	Deletion Referential Integrity: A book title cannot be deleted if physical copies exist. A physical copy cannot be removed if current status is Borrowed or Reserved.	Database Integrity Standard & Catalog Policy §6.1
BR-10	Separation of Roles: Readers have read-only access to catalog/personal data; Librarians manage circulation and catalog; Administrators manage user credentials.	Security Access Policy §1.3

Acceptance Criteria

The acceptance criteria define verifiable conditions of satisfaction for each feature module in Given / When / Then format. They form the contractual baseline for acceptance test cases in PA4:

Feature F01: User Authentication & Role-Based Access Control
• AC-F01-01: Given a registered user with valid credentials, When they submit login credentials, Then the system authenticates the user, returns a valid JWT, and navigates to the role-specific landing page.
• AC-F01-02: Given invalid or mismatched credentials, When the user attempts to log in, Then the system displays a descriptive error message and prevents session creation.
• AC-F01-03: Given an authenticated user with Reader role, When they attempt to access /admin/users or staff circulation endpoints, Then the system denies access returning HTTP 403 Forbidden.

Feature F02: Book Catalog Management
• AC-F02-01: Given unique ISBN and complete metadata, When a Librarian creates a new book title, Then the record is successfully saved in the catalog and appears in search results.
• AC-F02-02: Given an existing ISBN in the catalog, When a Librarian attempts to create another title with that ISBN, Then the system rejects the operation with a duplicate ISBN error.
• AC-F02-03: Given a book title with physical copies registered, When a Librarian attempts deletion, Then the system blocks deletion and instructs removal of copies first (BR-09).

Feature F03: Book Copy Inventory Management
• AC-F03-01: Given a valid book title and unique Barcode ID, When a Librarian registers a physical copy, Then the copy is initialized with status Available and assigned to shelf location.
• AC-F03-02: Given a physical copy currently marked Borrowed, When a Librarian attempts deletion, Then the deletion is blocked with a state conflict error.

Feature F04: Book Search & Browse
• AC-F04-01: Given a search query matching a title or author keyword, When the user executes the search, Then the system returns matching book titles in <= 1.0s (NFR-01) with real-time available copy counts.
• AC-F04-02: Given active catalog filters (e.g. Category = Computer Science), When filtered, Then only books matching the criteria are rendered.

Feature F05: Borrow & Return Management (Circulation Desk)
• AC-F05-01: Given a Reader with < 3 active loans and 0 overdue loans, When a Librarian checks out an Available copy, Then a loan record is created with due date = today + 14 days, and copy status becomes Borrowed (BR-01, BR-02).
• AC-F05-02: Given a Reader with 3 active loans (BR-01) or at least 1 overdue loan (BR-04), When checkout is initiated, Then the system rejects checkout and displays the policy block reason.
• AC-F05-03: Given a borrowed copy returned with no pending reservations, When check-in is confirmed, Then loan status changes to Returned, return date is logged, and copy status becomes Available.
• AC-F05-04: Given a returned copy of a book title with pending reservations, When check-in is confirmed, Then copy transitions to Reserved and hold pickup window starts (72h) for the top FIFO reader (BR-08).
• AC-F05-05: Given a current active loan that is not overdue and has no title reservations, When renewal is requested, Then due date is extended by 14 days (maximum 1 renewal, BR-03).

Feature F06: Book Reservation (Hold Processing)
• AC-F06-01: Given a book title with 0 Available copies, When an eligible Reader confirms reservation, Then the reservation is recorded at the end of the FIFO queue (BR-05, BR-07).
• AC-F06-02: Given a book title with >= 1 Available copy, When a Reader attempts to place a reservation, Then the system rejects reservation and prompts them to borrow the available copy.
• AC-F06-03: Given a held copy not collected within 72 hours (BR-08), When hold expiration is processed, Then reservation status changes to Expired and copy routes to the next FIFO patron or becomes Available.

Feature F07: Reader Overdue & Loan Monitoring Dashboard
• AC-F07-01: Given a logged-in Reader, When viewing their dashboard, Then all active loans,

due dates, overdue warnings, and active reservation statuses are rendered accurately.
• AC-F07-02: Given overdue loans in the system, When a Librarian views the overdue dashboard, Then all overdue loans are sorted by days overdue with reader contact details.

Use Case Model

Use Case Diagram

The UniLib use-case model captures role-to-goal interactions across Reader, Librarian, and Administrator actors. The full use-case diagram is maintained in the system design baseline, mapping actors to use cases UC-01 through UC-12.

Actors

Actor	Description
Reader	University student or faculty member who searches/browses catalog titles, views copy availability, places book reservations when copies are exhausted, and tracks personal borrowing records.
Librarian	Circulation staff member who manages book catalog metadata, tracks physical copy inventory, processes book checkouts and returns at the circulation desk, and manages hold shelf reservations.
Administrator	System administrator who provisions and manages user accounts, assigns role permissions, maintains system integrity, and monitors operational system health.

Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Log In & Manage Session	Reader, Librarian, Admin	Must*	FR-F01-01, FR-F01-02, FR-F01-03
UC-02	Manage User Accounts	Administrator	Must	FR-F01-04
UC-03	Manage Book Catalog (CRUD Titles)	Librarian	Must*	FR-F02-01, FR-F02-02, FR-F02-03, FR-F02-04
UC-04	Manage Book Copies & Inventory	Librarian	Must	FR-F03-01, FR-F03-02, FR-F03-03, FR-F03-04
UC-05	Search & Browse Book Catalog	Reader, Librarian	Must*	FR-F04-01, FR-F04-02, FR-F04-03, FR-F04-04
UC-06	View Book Details & Copy Availability	Reader, Librarian	Must	FR-F04-03, FR-F02-04
UC-07	Borrow Book (Circulation Checkout)	Librarian	Must*	FR-F05-01, FR-F05-02, FR-F05-03

UC-08	Return Book (Circulation Check-in)	Librarian	Must*	FR-F05-04, FR-F05-05
UC-09	Renew Book Loan	Reader, Librarian	Must	FR-F05-06, BR-03
UC-10	Reserve Book Title (Place Hold)	Reader	Must*	FR-F06-01, FR-F06-02, FR-F06-03
UC-11	Cancel Reservation & Process Expiry	Reader, System / Librarian	Must	FR-F06-04, FR-F06-05, BR-08
UC-12	View Reader Dashboard & Overdue Desk	Reader, Librarian	Should	FR-F07-01, FR-F07-02, FR-F07-03

Use Case Specifications

The core Must-have use cases (along with key circulation workflows) are specified below using the standard template format with numbered main, alternative, and exception flows:

Use Case UC-01

Use case ID	UC-01
Use case name	Log In & Manage Session
Brief description	Authenticate an authorized user via username/email and password credentials and establish a secure role-based session token.
Primary actor	Reader, Librarian, Administrator
Secondary actors	Authentication Service / Database Service
Priority	Must*
Related requirements	FR-F01-01, FR-F01-02, FR-F01-03
Trigger	User navigates to the Login page (SCR-01) or attempts to access a protected URL.
Pre-conditions	User account has been registered by the Administrator and is currently in Active status.
Post-conditions	A cryptographically signed JWT with role claim is issued; user is redirected to their role landing dashboard; or session is terminated upon logout.
Main flow	1. User opens the login interface (SCR-01). 2. User enters username/email and password, then submits credentials. 3. System validates input formats and hashes password with bcrypt (salt >= 10). 4. System verifies credentials against the user database and checks active status. 5. System generates signed JWT containing user ID and role claim. 6. System redirects user to their role landing page (Reader: SCR-11, Librarian: SCR-07, Admin: SCR-02).
Alternative flows	A1 (at step 2): User toggles password visibility. A2 (at step 6): User clicks Logout; system clears client session token and redirects to SCR-01.
Exception flows	E1 (at step 4): Invalid username or password; system displays "Invalid credentials" error banner without revealing which field failed (NFR-03). E2 (at step 4): Account is deactivated; system displays "Account suspended, contact Administrator" (FR-F01-04).
Business rules	BR-10 (Separation of Roles)
Non-functional constraints	NFR-03 (Password Hashing bcrypt >= 10), NFR-04 (JWT Authorization, 401/403 handling)

Use Case UC-03

Use case ID	UC-03
Use case name	Manage Book Catalog (CRUD Titles)
Brief description	Create, update, and manage bibliographic book title records in the catalog while maintaining referential integrity.
Primary actor	Librarian
Secondary actors	Database Service
Priority	Must*
Related requirements	FR-F02-01, FR-F02-02, FR-F02-03, FR-F02-04
Trigger	Librarian accesses Catalog Management Console (SCR-03).
Pre-conditions	Authenticated user with Librarian role.
Post-conditions	Book title record is created or updated in the catalog database; copy statistics recalculated.
Main flow	1. Librarian opens Catalog Management (SCR-03). 2. Librarian clicks "Add New Book" and enters ISBN, Title, Author(s), Category, Publisher, and Publication Year. 3. System validates ISBN uniqueness and metadata field formats. 4. System persists new book title and renders updated catalog table with copy count badge = 0. 5. Librarian may edit metadata of existing titles or inspect copy statistics.
Alternative flows	A1 (at step 2): Librarian edits existing title metadata; system persists changes and updates display. A2 (at step 5): Librarian requests title deletion; system verifies 0 physical copies and 0 active holds, then removes record.
Exception flows	E1 (at step 3): ISBN already exists; system rejects submission with duplicate ISBN error (FR-F02-01). E2 (at step A2): Title has linked physical copies or active loans/holds; system blocks deletion per BR-09.
Business rules	BR-09 (Deletion Referential Integrity)
Non-functional constraints	NFR-04 (Librarian RBAC Authorization), NFR-08 (Reliable persistence)

Use Case UC-05

Use case ID	UC-05
Use case name	Search & Browse Book Catalog
Brief description	Search the library catalog by keywords and filter by category or availability status with real-time copy counts.

Primary actor	Reader, Librarian
Secondary actors	Search Indexer / Database Service
Priority	Must*
Related requirements	FR-F04-01, FR-F04-02, FR-F04-03, FR-F04-04
Trigger	User navigates to Catalog Search view (SCR-05) or enters search terms in header bar.
Pre-conditions	System catalog is initialized and available.
Post-conditions	Matching book titles are displayed with real-time available copy counts and shelf locations.
Main flow	1. User accesses Search & Browse interface (SCR-05). 2. User enters search keywords (matching Title, Author, Category, or ISBN). 3. User applies optional filters (Category, Availability: "Available Now") and sorting (Title A-Z, Year). 4. System queries catalog index and returns paginated results (default 10 items/page) within <= 1.0s (NFR-01). 5. System renders book cards showing cover, bibliographic metadata, shelf location, and available copy counter. 6. User clicks a book card to view detailed copy inventory (SCR-06).
Alternative flows	A1 (at step 2): User leaves search bar empty; system displays paginated list of all active catalog titles. A2 (at step 3): User switches pagination page; system fetches next batch via server-side pagination.
Exception flows	E1 (at step 4): No catalog titles match search query; system displays friendly "No books found" message with filter reset suggestion.
Business rules	BR-10 (Public / Reader Access)
Non-functional constraints	NFR-01 (Search Latency <= 1.0s at P95 under 100 VU across 10,000 records), NFR-02 (LCP <= 2.5s)

Use Case UC-07

Use case ID	UC-07
Use case name	Borrow Book (Circulation Checkout)
Brief description	Librarian issues a book loan to an eligible Reader, calculating due date and transitioning copy to Borrowed.
Primary actor	Librarian
Secondary actors	Database Service
Priority	Must*
Related requirements	FR-F05-01, FR-F05-02, FR-F05-03, FR-F06-06

Trigger	Reader presents library card/ID and physical book copy at the circulation desk (SCR-07).
Pre-conditions	Authenticated Librarian; Reader account is active; copy is physically present.
Post-conditions	Active loan record created with due date = today + 14 days; copy status updated to Borrowed.
Main flow	1. Librarian opens Circulation Desk - Borrow (SCR-07). 2. Librarian enters Reader ID; system retrieves patron profile and verifies eligibility: < 3 active loans (BR-01) and 0 overdue loans (BR-04). 3. Librarian enters/scans Copy Barcode ID; system verifies copy status is Available. 4. System locks copy row with SELECT ... FOR UPDATE (NFR-05) within a database transaction. 5. System computes due date = checkout date + 14 calendar days (BR-02). 6. System creates active Loan record and transitions Copy status to Borrowed. 7. System displays checkout confirmation banner with due date and updated patron loan count.
Alternative flows	A1 (at step 3): Copy is currently Reserved on hold shelf for this specific Reader; system fulfills reservation (FR-F06-06) and converts hold into active loan.
Exception flows	E1 (at step 2): Reader has 3 active loans; system blocks checkout with "Maximum loan limit (3 books) reached" per BR-01. E2 (at step 2): Reader has >= 1 overdue loan; system blocks checkout with "Borrowing suspended: overdue loan present" per BR-04. E3 (at step 3): Copy status is Borrowed or Damaged; system blocks checkout with state conflict error.
Business rules	BR-01 (Max 3 Loans), BR-02 (14-day Duration), BR-04 (Overdue Suspension), BR-10 (Librarian RBAC)
Non-functional constraints	NFR-05 (Transactional Atomic Checkout, 0 Race Conditions)

Use Case UC-08

Use case ID	UC-08
Use case name	Return Book (Circulation Check-in)
Brief description	Librarian processes a returned book copy, recording check-in timestamp, assessing condition, and routing to hold shelf or available.
Primary actor	Librarian
Secondary actors	Database Service / Reservation Scheduler
Priority	Must*
Related requirements	FR-F05-04, FR-F05-05
Trigger	Patron returns physical book copy to circulation desk (SCR-08).
Pre-conditions	Authenticated Librarian; copy was checked out on an active loan.
Post-conditions	Loan status marked Returned; copy status transitions to Reserved (if holds exist) or Available.

Main flow	1. Librarian opens Circulation Desk - Return (SCR-08). 2. Librarian enters/scans Copy Barcode ID. 3. System locates active Loan record and locks copy row in database transaction (NFR-05). 4. System records return timestamp and marks Loan status as Returned. 5. System evaluates whether pending reservations exist for the book title in FIFO queue (BR-07). 6. If zero pending reservations: system transitions copy status to Available and directs shelving. 7. System displays check-in confirmation toast with shelving destination.
Alternative flows	A1 (at step 5): Pending reservation exists; system transitions copy status to Reserved, allocates copy to top FIFO patron, initiates 72-hour pickup window (BR-08), and displays hold routing notice.
Exception flows	E1 (at step 2): Copy ID does not correspond to an active loan; system alerts Librarian "No active loan found for this copy". E2 (at step 4): Copy is returned in damaged condition; Librarian flags condition; copy transitions to Damaged/Lost (never routed to hold shelf per TA-FB-02) and pending reservation remains in queue.
Business rules	BR-07 (FIFO Queue), BR-08 (72h Hold Window), BR-09 (Integrity)
Non-functional constraints	NFR-05 (Atomic Check-in and Hold Routing)

Use Case UC-10

Use case ID	UC-10
Use case name	Reserve Book Title (Place Hold)
Brief description	Reader places a title-level reservation when zero copies are available, entering the FIFO queue.
Primary actor	Reader
Secondary actors	Reservation Queue Service
Priority	Must*
Related requirements	FR-F06-01, FR-F06-02, FR-F06-03
Trigger	Reader clicks "Reserve Book" button on Book Detail view (SCR-06).
Pre-conditions	Authenticated Reader; zero copies of book title are Available (BR-05); Reader has < 2 active reservations (BR-06); Reader has 0 overdue loans (BR-04) and no active loan of this title.
Post-conditions	Reservation record created with timestamp and queue position; reader informed of hold policy.

Main flow	1. Reader views Book Detail (SCR-06) showing 0 Available copies. 2. Reader clicks "Reserve Book" action button. 3. System displays Reservation Confirmation Modal (SCR-09) showing estimated queue position and 72-hour pickup policy disclaimer. 4. Reader confirms reservation request. 5. System verifies reservation constraints in database transaction: < 2 active holds, no active loan for title, no overdue items. 6. System appends reservation to title FIFO queue with creation timestamp (BR-07). 7. System displays success alert and updates Reader Personal Dashboard (SCR-11).
Alternative flows	A1 (at step 4): Reader cancels modal; system closes modal without recording hold.
Exception flows	E1 (at step 1): At least 1 copy is Available; system disables Reserve button and prompts reader to borrow available copy (BR-05). E2 (at step 5): Reader already has 2 active reservations; system rejects request with "Maximum active reservations (2) reached" (BR-06). E3 (at step 5): Reader currently has overdue loan; system blocks reservation per BR-04.
Business rules	BR-04 (Overdue Suspension), BR-05 (Reservation Eligibility), BR-06 (Max 2 Holds), BR-07 (FIFO Queue)
Non-functional constraints	NFR-04 (Reader RBAC), NFR-05 (Atomic Hold Allocation)

Use Case UC-12

Use case ID	UC-12
Use case name	View Reader Dashboard & Overdue Desk
Brief description	Reader views personal borrowings and holds; Librarian monitors overdue loans across the library.
Primary actor	Reader, Librarian
Secondary actors	Database Service
Priority	Should
Related requirements	FR-F07-01, FR-F07-02, FR-F07-03
Trigger	User navigates to My Dashboard (SCR-11) or Overdue Monitoring Console (SCR-12).
Pre-conditions	Authenticated user with Reader or Librarian role.
Post-conditions	Role-appropriate circulation summaries, due dates, overdue badges, and action triggers rendered.

Main flow	1. Authenticated user navigates to their circulation monitoring view. 2. System queries active loans, calculated due dates, and pending reservations for the user or library. 3. For Reader (SCR-11): system renders active loans table, due-date progress bars, eligible Renew button, active holds list, and prominent Overdue warning banner if overdue items exist (FR-F07-02). 4. For Librarian (SCR-12): system renders delinquent loans table sorted by days overdue with reader contact details and copy IDs (FR-F07-03). 5. User reviews records or initiates eligible actions (e.g. renewal or patron contact).
Alternative flows	A1 (at step 4): Librarian filters overdue list by days overdue (> 7 days, > 14 days) or exports summary view.
Exception flows	E1 (at step 2): Database/network timeout occurs; system displays user-friendly error banner without crashing (NFR-08).
Business rules	BR-03 (Renewal Policy), BR-04 (Overdue Suspension), BR-10 (Separation of Roles)
Non-functional constraints	NFR-04 (RBAC Protection), NFR-06 (Accessibility, WCAG Contrast >= 4.5:1), NFR-08 (Graceful Error Handling)

Prototype / Mockups

The screen inventory below defines the 12 core user interface views corresponding to the system use cases. The interactive Figma prototype is available at https://www.figma.com/design/VGG2HSwsJmU0uNHUAjmvdZ/hehe with accessibility compliance verified per NFR-06:

Screen	Related use case(s)	Mockup / link
SCR-01 Authentication & Login Screen	UC-01	Figma Prototype: /wireframes/SCR-01-Login (docs/design/wireframes.md)
SCR-02 Admin User Management Screen	UC-02	Figma Prototype: /wireframes/SCR-02-UserManagement
SCR-03 Librarian Book Catalog Management Screen	UC-03	Figma Prototype: /wireframes/SCR-03-CatalogCRUD
SCR-04 Book Copy Inventory Management Screen	UC-04	Figma Prototype: /wireframes/SCR-04-CopyInventory
SCR-05 Catalog Search & Browse Screen	UC-05	Figma Prototype: /wireframes/SCR-05-SearchBrowse
SCR-06 Book Title Detail & Availability Screen	UC-06, UC-10	Figma Prototype: /wireframes/SCR-06-BookDetail
SCR-07 Circulation Desk - Borrow (Checkout) Screen	UC-07	Figma Prototype: /wireframes/SCR-07-BorrowCheckout
SCR-08 Circulation Desk - Return (Check-in) Screen	UC-08	Figma Prototype: /wireframes/SCR-08-ReturnCheckin
SCR-09 Book Reservation Modal / Confirmation View	UC-10	Figma Prototype: /wireframes/SCR-09-ReserveModal
SCR-10 Librarian Hold Shelf & Reservation Queue Screen	UC-10, UC-11	Figma Prototype: /wireframes/SCR-10-HoldQueue
SCR-11 Reader Personal Dashboard Screen	UC-09, UC-11, UC-12	Figma Prototype: /wireframes/SCR-11-ReaderDashboard
SCR-12 Librarian Overdue & Circulation Monitoring Screen	UC-12	Figma Prototype: /wireframes/SCR-12-OverdueDesk

Requirements Traceability Matrix

The forward traceability matrix establishes links from functional requirements to use cases, wireframe screens, PA3 architecture design elements, and PA4 test cases:

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-F01-01	UC-01	SCR-01	[TBD: PA3 AuthController / POST /auth/login]	[TBD: PA4 TC-F01-01]
FR-F01-02	UC-01	SCR-01	[TBD: PA3 AuthMiddleware / JWT RBAC]	[TBD: PA4 TC-F01-02]
FR-F01-04	UC-02	SCR-02	[TBD: PA3 UserController / REST endpoints]	[TBD: PA4 TC-F01-04]
FR-F02-01	UC-03	SCR-03	[TBD: PA3 BookCatalogController / POST /books]	[TBD: PA4 TC-F02-01]
FR-F02-03	UC-03	SCR-03	[TBD: PA3 BookCatalogController / DELETE /books/:id]	[TBD: PA4 TC-F02-03]
FR-F03-01	UC-04	SCR-04	[TBD: PA3 BookCopyController / POST /copies]	[TBD: PA4 TC-F03-01]
FR-F03-03	UC-04	SCR-04	[TBD: PA3 BookCopy state machine schema]	[TBD: PA4 TC-F03-03]
FR-F04-01	UC-05	SCR-05	[TBD: PA3 SearchController / GET /books/search]	[TBD: PA4 TC-F04-01]
FR-F04-03	UC-06	SCR-06	[TBD: PA3 BookDetailController / GET /books/:id]	[TBD: PA4 TC-F04-03]
FR-F05-01	UC-07	SCR-07	[TBD: PA3 LoanController / POST /loans/checkout]	[TBD: PA4 TC-F05-01]
FR-F05-04	UC-08	SCR-08	[TBD: PA3 LoanController / POST /loans/return]	[TBD: PA4 TC-F05-04]
FR-F05-06	UC-09	SCR-11	[TBD: PA3 LoanController / POST /loans/:id/renew]	[TBD: PA4 TC-F05-06]
FR-F06-01	UC-10	SCR-09	[TBD: PA3 ReservationController / POST /reservations]	[TBD: PA4 TC-F06-01]
FR-F06-04	UC-11	SCR-10	[TBD: PA3 Reservation hold timer & routing]	[TBD: PA4 TC-F06-04]
FR-F07-01	UC-12	SCR-11	[TBD: PA3 DashboardController / GET /reader/loans]	[TBD: PA4 TC-F07-01]
FR-F07-03	UC-12	SCR-12	[TBD: PA3 OverdueController / GET /librarian/overdue]	[TBD: PA4 TC-F07-03]

Requirements Review

The formal requirements review was conducted through TA Consultation (SES-01 on 2026-11-01) and Team Walkthrough (SES-02 on 2026-11-02). Identified issues were resolved and approved to establish the frozen requirements baseline:

#	Issue found	Requirement(s)	Resolution
1	Undefined reader loan limit and borrowing period in PA1 proposal baseline	FR-F05-02, BR-01, BR-02	Resolved during PA2 Kickoff (Decision D-PA2-01): Capped at 3 concurrent books per reader and 14 calendar days borrowing duration with max 1 renewal.
2	Undefined reservation hold pickup window duration upon copy return	FR-F06-04, BR-08	Resolved during PA2 Kickoff (Decision D-PA2-02): Fixed hold pickup window at 72 hours (3 calendar days) before automated queue advancement or release.
3	Potential concurrency race conditions during simultaneous checkouts of last copy or hold routing	FR-F05-01, FR-F06-04, NFR-05	Resolved during PA2 Kickoff (Decision D-PA2-03 & Q1 input): Mandated PostgreSQL ACID transactions with SELECT ... FOR UPDATE row-level locks.

AI Usage Log

In compliance with the course GenAI policy and Team Contract §3.6, all significant AI tool usage across team members is fully disclosed below:

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Phan Thế Vinh (Team Lead / PM · Tester)	Claude 3.5 Sonnet, ChatGPT (GPT-4o)	PA2 Kickoff, open items resolution (loan limit, pickup window), Master ID Registry freeze (WBS 2.1.1 / Task PA2-V1), Acceptance Criteria drafting (AC-F01-01 to AC-F07-02), and RTM schema setup.	Verified against PA0/PA1 scope and team contract; confirmed by all 4 members during kickoff meeting; reconciled 32 FRs, 8 NFRs, 10 BRs, 12 UCs, 12 SCRs, 30 ACs to prevent numbering overlaps and unblock parallel authoring.
2	Nguyễn Thanh Thảo Nguyên (Business Analyst)	ChatGPT (GPT-4o), Google Gemini 1.5 Pro	Domain entity scoping, Glossary draft (02-glossary.md), Problem Statement root cause analysis, all 32 Functional Requirements (F01–F07), 10 Business Rules (BR-01 to BR-10), and Use Case specifications.	Audited draft use cases against domain policies; eliminated hallucinated payment/e-book features; ensured checkout/return flows strictly enforce BR-01..BR-10; added explicit damaged copy handling in UC-08 per TA feedback; verified terminology against glossary.
3	Vũ Thị Ánh (Designer / Software Architect)	Claude 3.5 Sonnet, ChatGPT (GPT-4o)	UI wireframe design (SCR-01..12), screen inventory hierarchy, interactive prototype assembly, UC-12 spec, WCAG 2.1 AA accessibility guidelines, and responsive CSS layout parameters.	Verified frames against Master ID Registry; audited WCAG contrast ratio (>= 4.5:1); tested public Figma link in incognito mode; added missing UI error banners/toasts (overdue block, 72h hold confirmation); exported 12 final PNG screenshots.
4	Trần Hà Anh Quân (Software Developer)	Claude 3.5 Sonnet, GitHub Copilot	Repository skeleton audit, PostgreSQL concurrency and ACID transaction analysis, row-level locking strategy (SELECT FOR UPDATE) for circulation states, NFR measurability review, and CI verification.	Audited AI recommendations against PostgreSQL 16 transactional semantics; mandated pessimistic row locks for copy/hold contention; confirmed repository remains planning skeleton without premature code; documented findings in pa2-quan-technical-review.md.